

Introduction

The Customer Segmentation Project analyzes customer behavior and demographics to group customers into meaningful segments using Python and Power BI.

Objective

Identify customer groups, improve marketing strategies, increase retention, and support business decision-making.

Tools and Technologies

Python, Scikit-learn, Pandas, NumPy, Matplotlib, Power BI, Microsoft Excel, GitHub.

Methodology

Data collection, cleaning, preprocessing, K-Means clustering, dashboard creation, and business insight generation.

Key Features

Customer clustering, purchase behavior analysis, demographic analysis, and interactive dashboards.

Results

The project identified customer groups that support targeted marketing and improved business decisions.

Challenges

Missing data handling, preprocessing, cluster selection, and visualization.

Learning Outcomes

Hands-on experience with Python, Power BI, machine learning, and business analytics.

Future Enhancements

Customer churn prediction, recommendation systems, customer lifetime value analysis, and real-time dashboards.

Conclusion

The project demonstrates how customer segmentation helps businesses understand customers and make data-driven decisions.